

Handheld & Embedded Calculator

Production-ready handheld electronics - custom firmware, PCB design, and a 3D-printed enclosure.

ADRIANO ZAGAR

C++ / ARDUINO / ESP32 / KICAD / SOLIDWORKS

FIRMWARE Custom C++ computational logic

CONTROL ESP32 microcontroller

DISPLAY 4-pin LCD interface

HARDWARE 2-layer PCB / MT3608 / TP4056 / 3.7 V rechargeable battery

ENCLOSURE SOLIDWORKS snap-fit housing

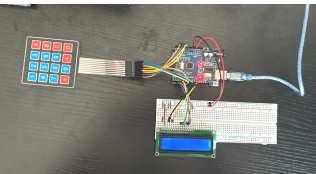
GOAL Compact, cost-efficient product



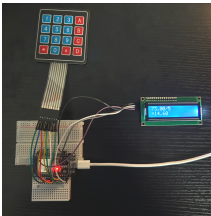
Final assembled handheld calculator

BUILD PROCESS

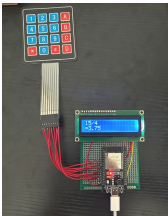
01	02	03	04	05	06
PROTOTYPE	MIGRATE	VALIDATE	SCHEMATIC	PCB	ENCLOSURE
Arduino Uno	ESP32 system	Perf-board	KiCad	2-layer board	SOLIDWORKS



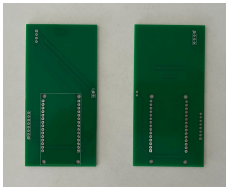
Arduino Uno Breadboard



ESP32 Breadboard



Perf-Board



Custom PCB

ENGINEERING HIGHLIGHTS

Cross-domain integration

Firmware, electronics, and enclosure

Spatial optimization

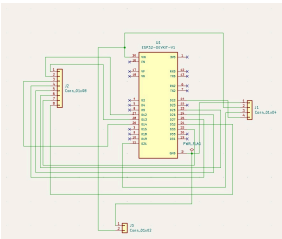
Compact handheld architecture

Design for manufacturing

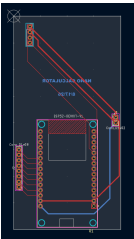
Efficient assembly and sourcing

Rechargeable power system

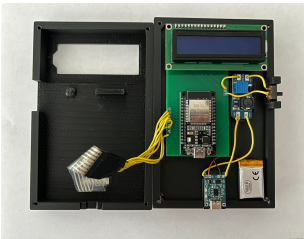
MT3608 / 3.7 V battery / TP4056



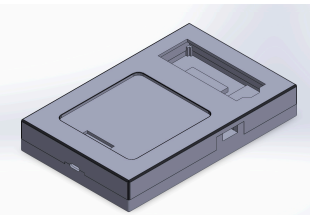
Schematic
KiCad



PCB Layout
KiCad



Internal Electronics
Rechargeable power system



Enclosure CAD
SOLIDWORKS